

DTU

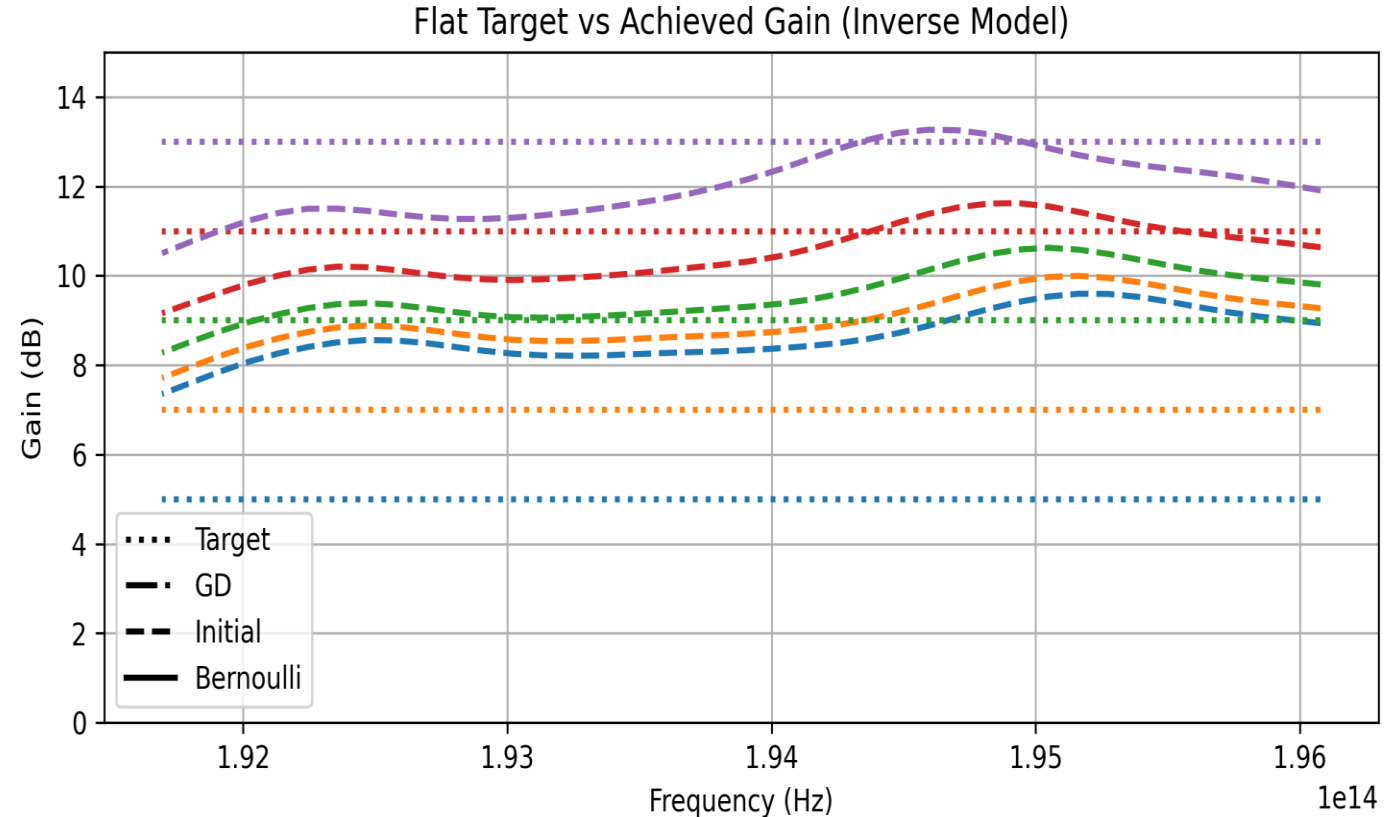


Reinforcement Learning Control of Raman Amplifiers

08.01.2026

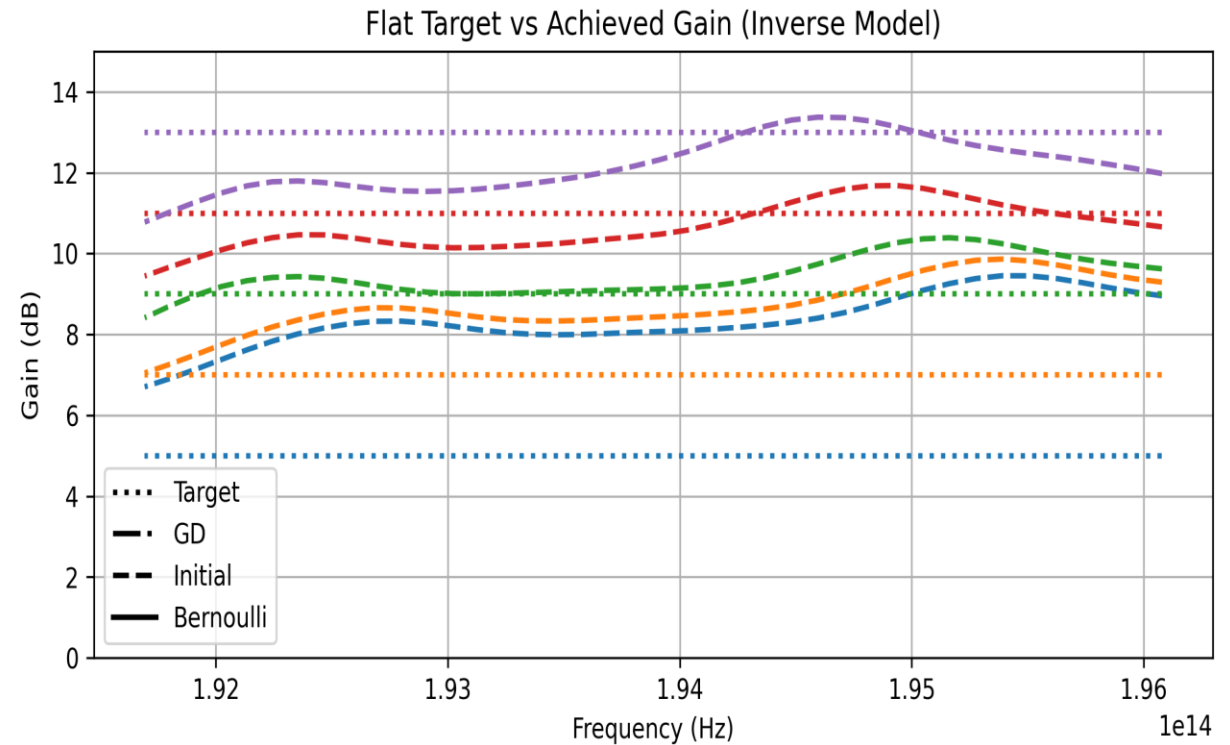
Inverse model

- The inverse model uses Random Projection Method/Extreme Learning to capture the inverse mapping of the non-linear system.
- It takes the output spectrum as input, and outputs the powers and wavelengths needed to achieve the desired spectrum.
- We can see that this inverse model outputs give the results which are biased towards the mean of the dataset.



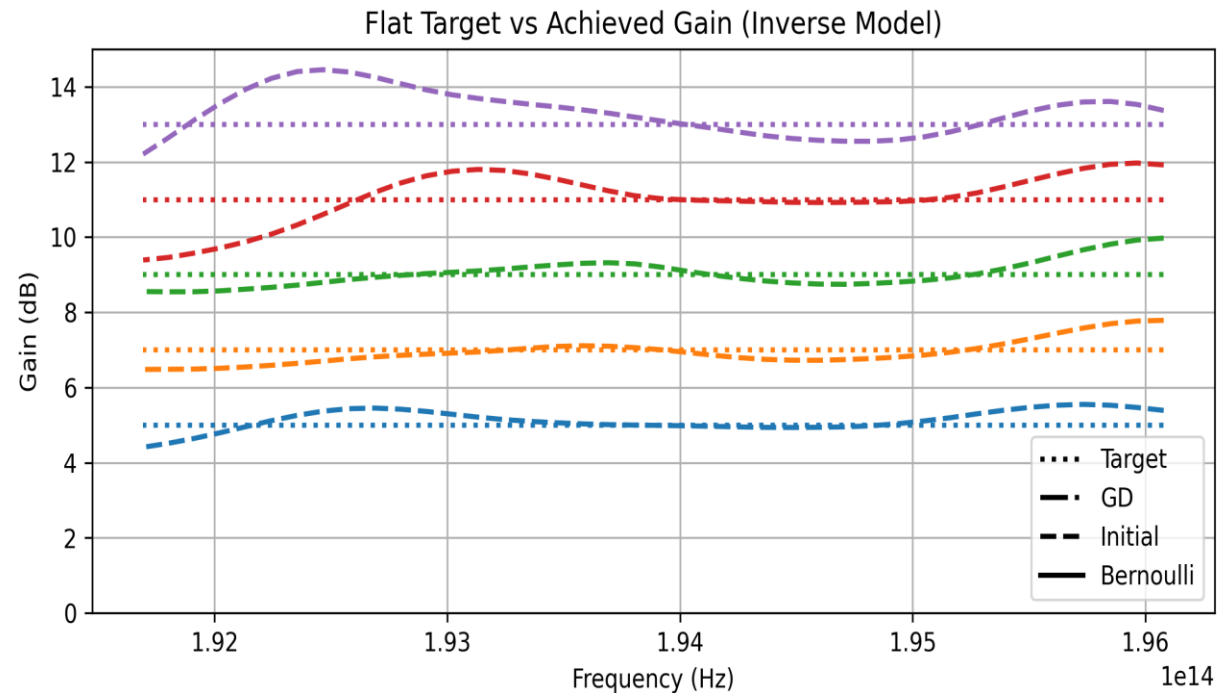
Inverse model

Slightly better performance of the Inverse model was achieved when sampling all models in the ensemble and using the forward model to select the best one instead of using the mean of all outputs



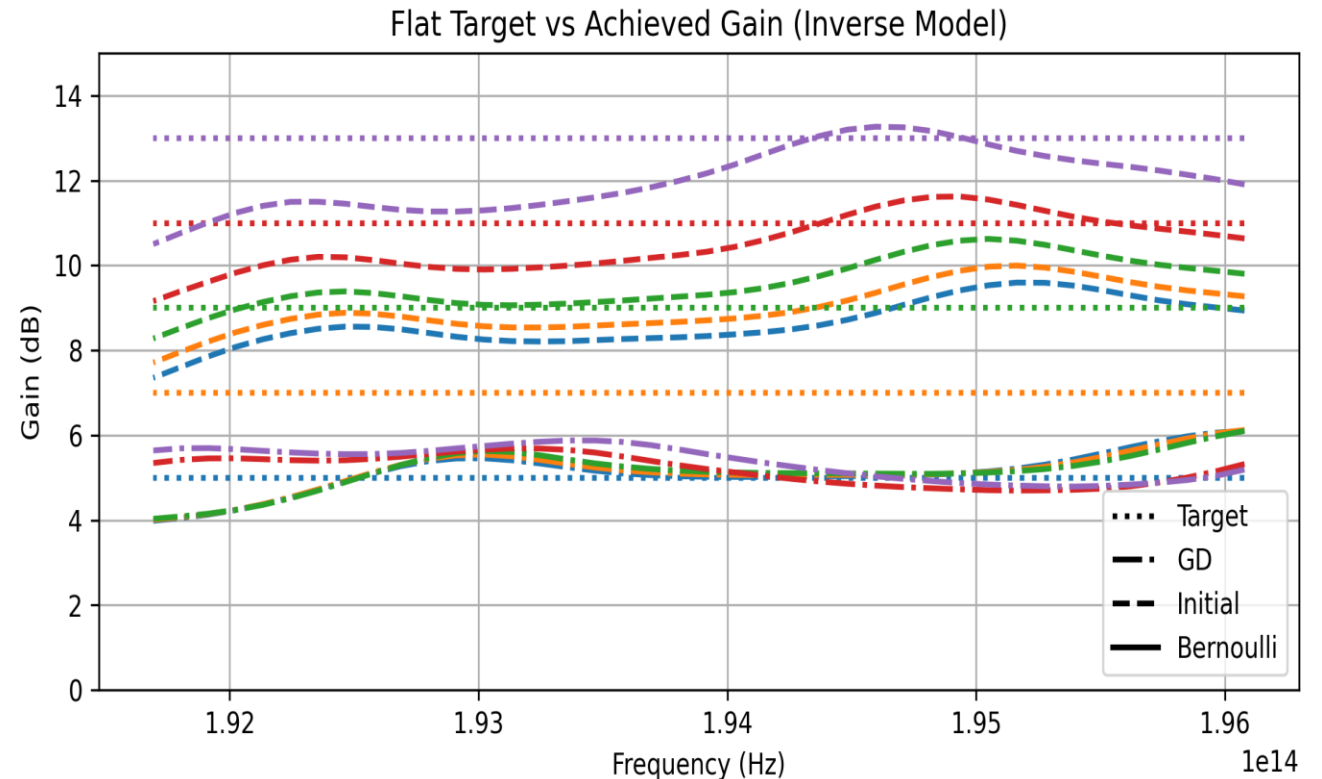
Update: Inverse model

- Further experimentation with the hyper-parameters resulted in a much better performance of the inverse model
- Now it achieves the error of:
 - Mean: 0.471 dB
 - Std: 0.308 dB
- This error statistic is close to the one achieved in the JLT paper (mean: 0.485 dB, std: 0.262 dB)



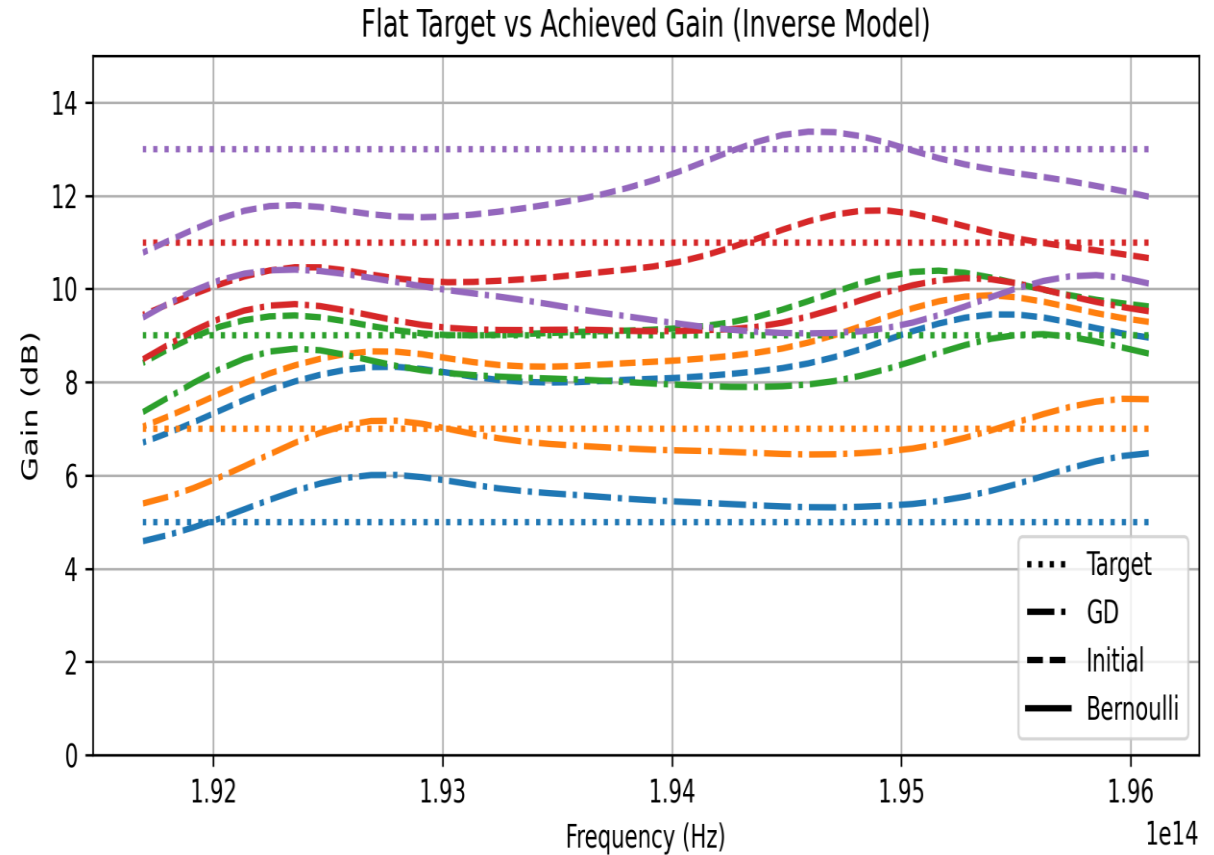
Inverse model + Gradient Descent Optimization

- In theory using the gradient descent based on the forward model should reduce the error, but in this case that doesn't happen.
- The behavior that is observed from applying gradient descent is that the output gets constantly reduced. Further inspection indicates that this trend is logarithmic.



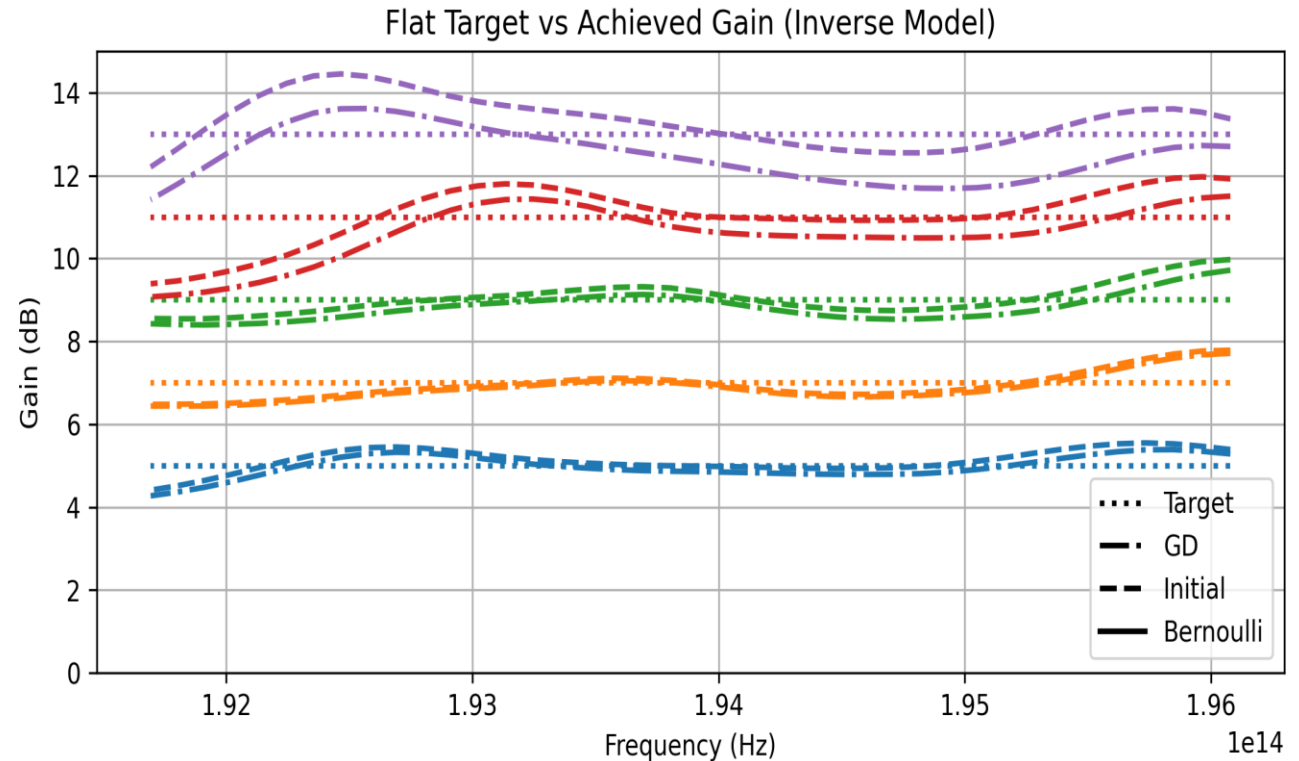
Inverse model + Gradient Descent Optimization

- Much better performance was obtained in the lower gains by introducing a stopping criterium to the control loop.
- The stopping criterium used was running the gradient descent optimization until the error started increasing.
- Still the gradient always points to low power side of the domain, meaning that gain spectra higher than the initial guess are not optimized.

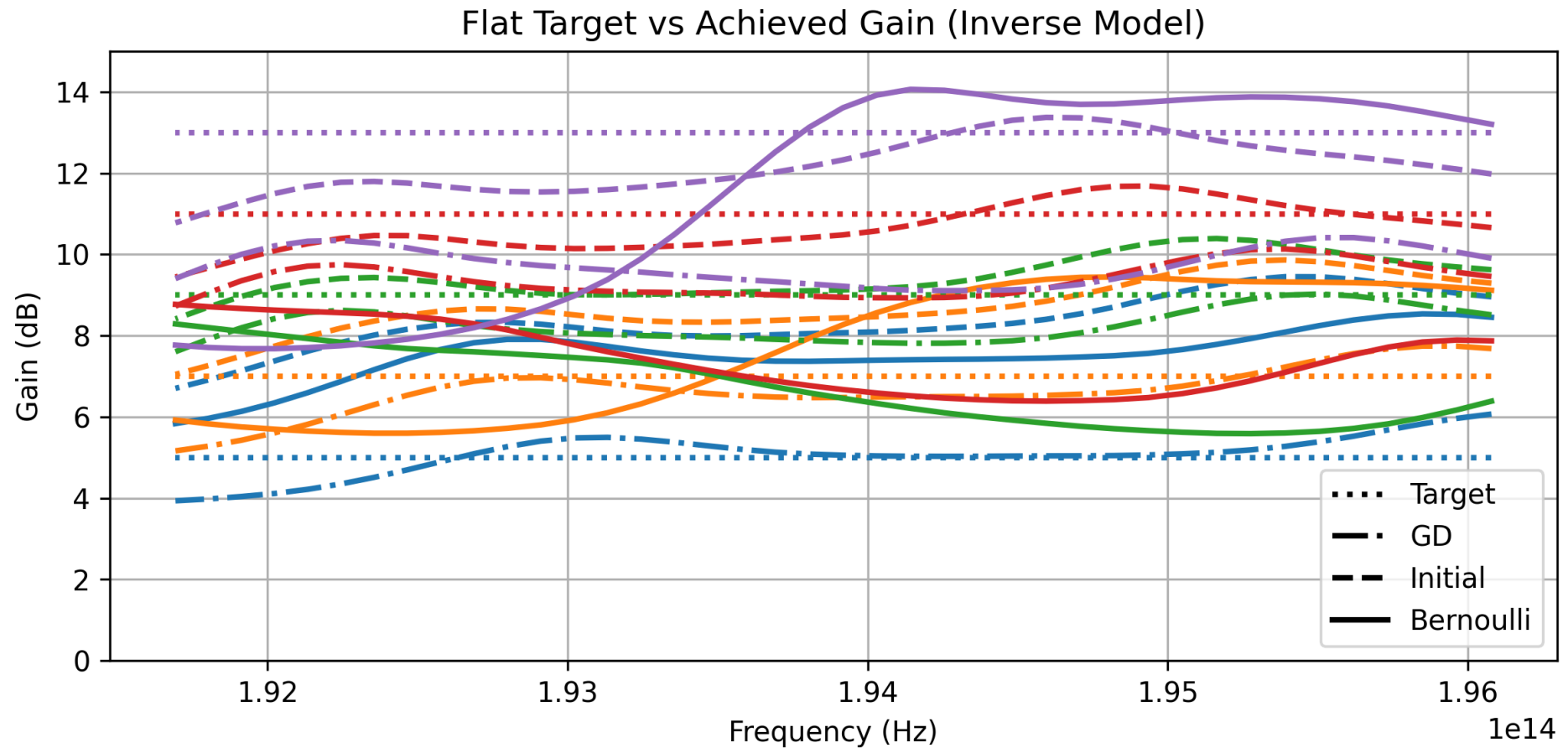


Updated: Inverse model + Gradient Descent Optimization

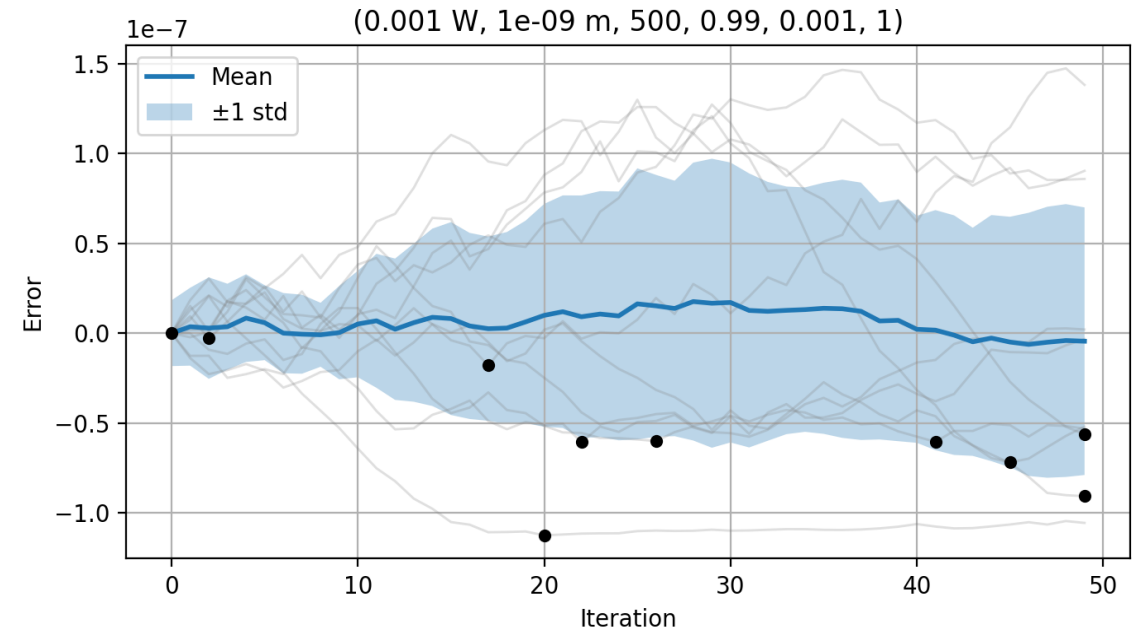
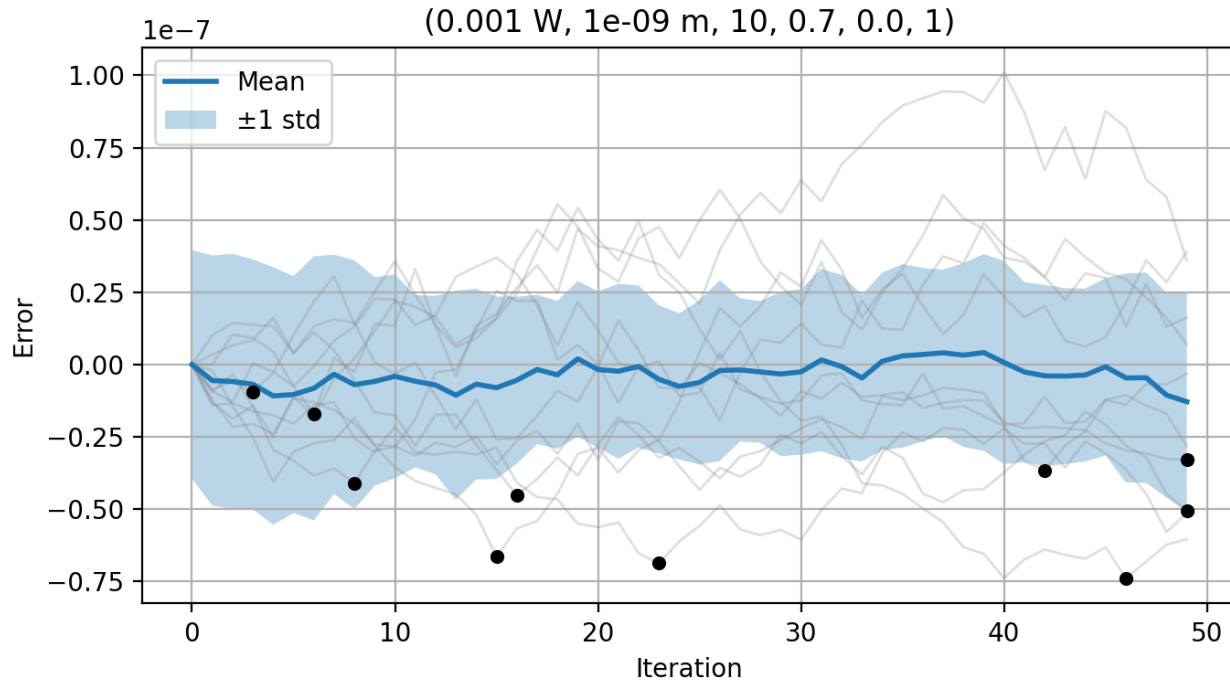
- With the better performing inverse model the gradient descent optimization doesn't have to optimize the output much, granting much better overall result
- The gradient descent controller still only controls the powers downwards, which only works if the initial guess is higher than the target.



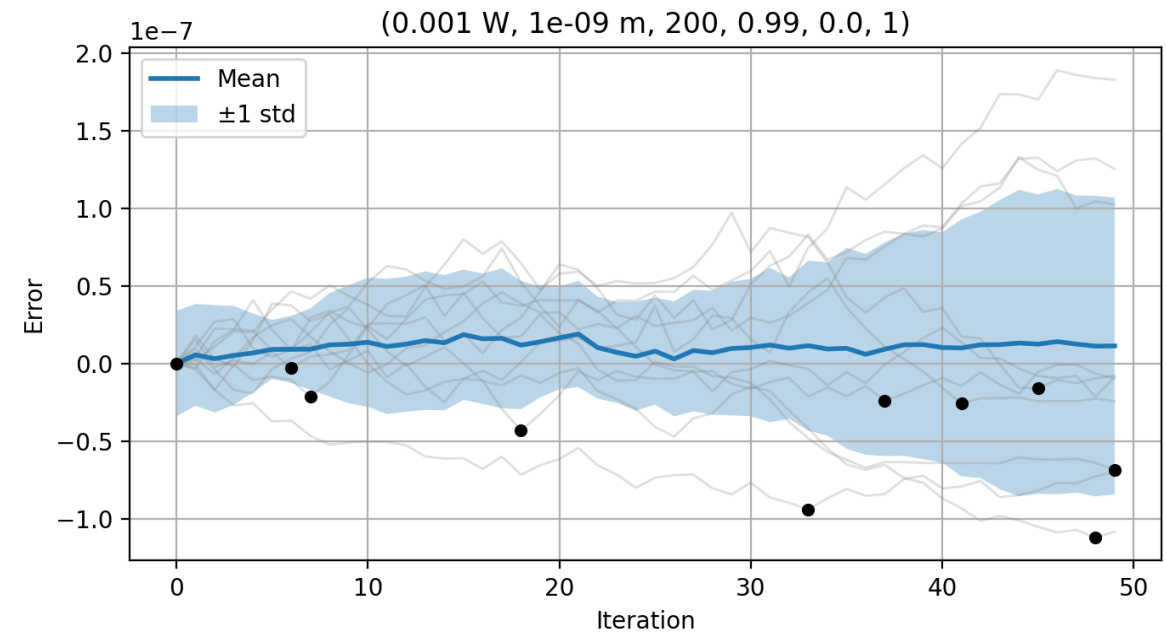
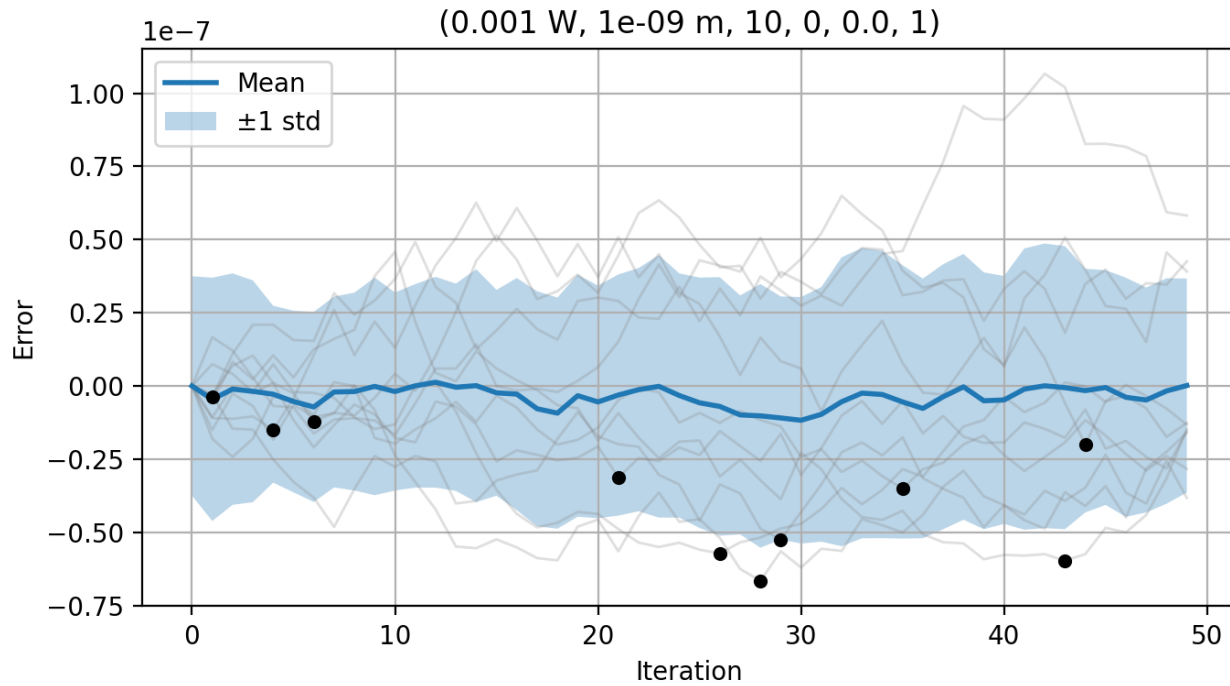
Inverse model + RL Optimization



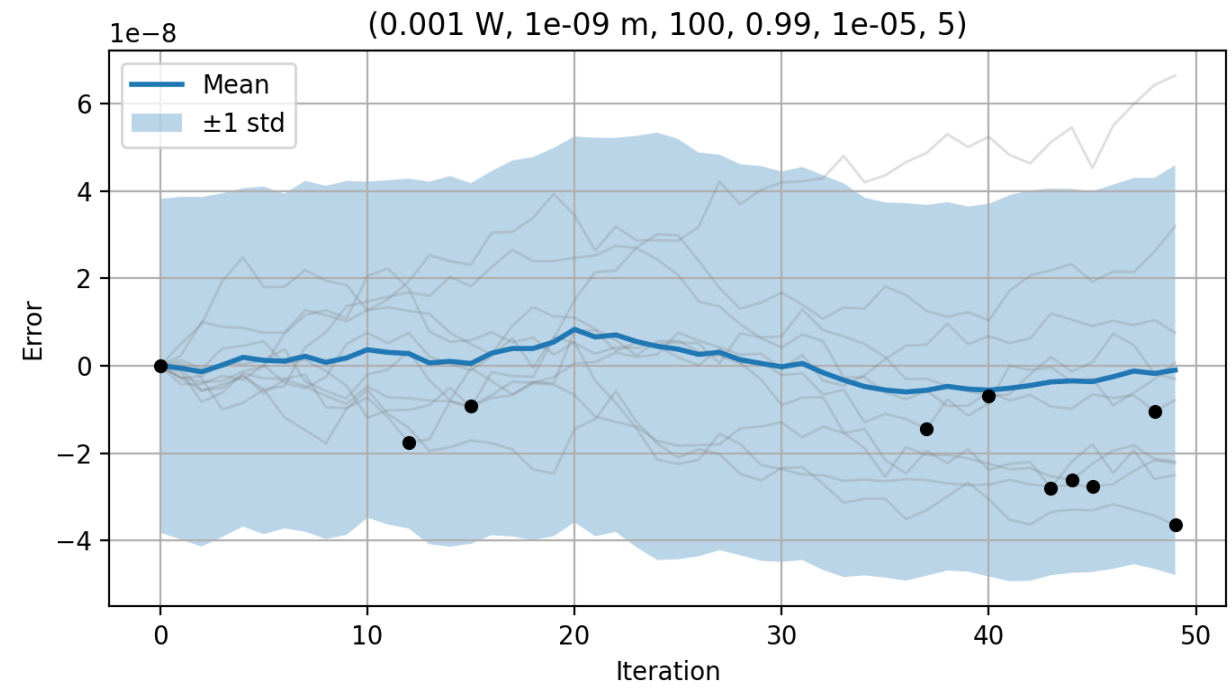
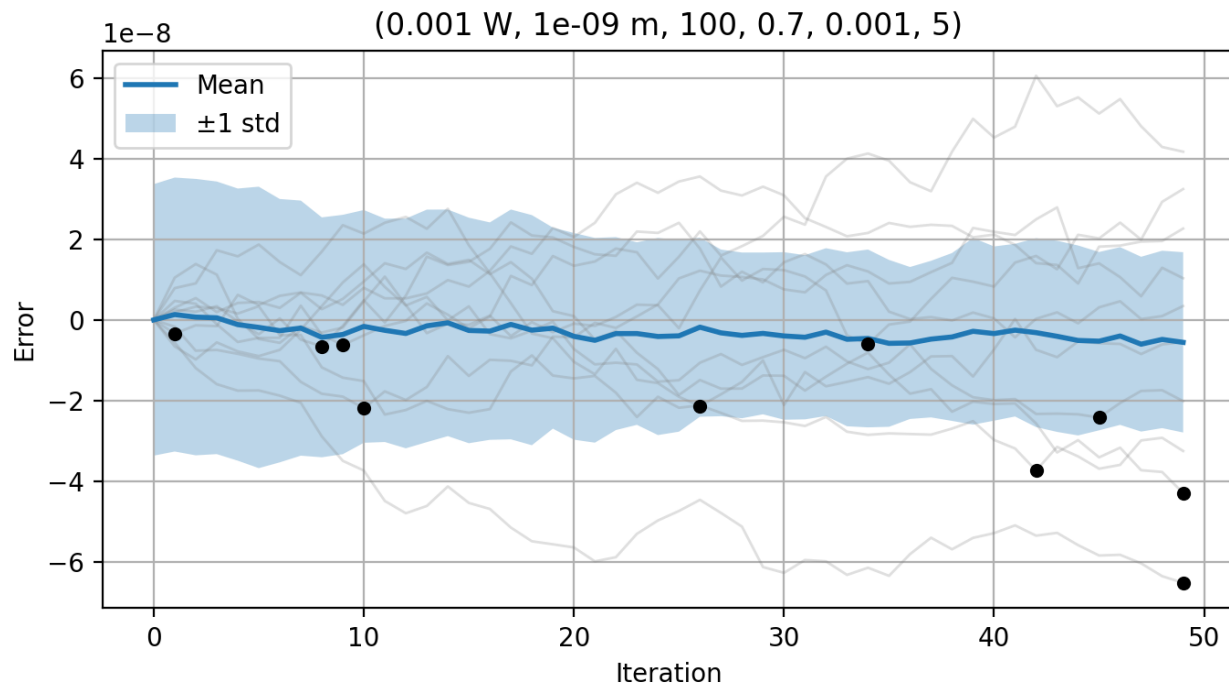
RL Controller MSE Loss



RL Controller MSE Loss



RL Controller MSE Loss



RL Controller - Potential Improvements

- Hyper-parameter tuning of Bernoulli controller didn't result in better performance so far.
- Optimizing hyper-parameters for much simpler problems proved quite difficult as well, so it could be a matter of persistence.
- Alternatively, since the Bernoulli controller does a good job at exploring the input space, it could use memory of the best set of parameters, which could be reproduced upon the end of optimization.

